College of Engineering and Technology

Project Work – Student Hand Book

Project Batch ID: B181

Degree/ program	B.Tech	**Specialisation**	Computer Science & Engineering
Academic Year	2025-2026 (Even)	**Semester**	6

Name of student	**Register Number**	**Department**	**Mobile Number**	**Email ID**
Akshat Gupta	RA2311003010021	C.TECH.	9289231500	ag6341@srmist.edu.in
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Title of the Project:	APMS v3.0: A Decentralized Smart Parking Framework with Edge AI and Federated LearningX
Project Site / Location	SRM IST, Kattankulathur, Chengalpattu District – 603203
Name and address of the company / organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
Name	Dr. Harriet Linda C		
Designation	Assistant Professor		
Department	Computing Technologies		
Campus	Kattankulathur		
Telephone	9486967281		
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Course Code	21CSP302L	Course Title	Project

Mission Statement

To develop a decentralized AI-powered smart parking management system that improves urban parking efficiency using Edge AI, Federated Learning, Blockchain, and Multi-Agent Reinforcement Learning while ensuring privacy, reduced congestion, financial transparency, and real-time vehicle management.

Problem (or) Product Description:

Urban parking congestion causes major traffic delays, fuel wastage, pollution, and financial inefficiencies. Traditional parking systems rely heavily on manual ticketing, centralized cloud processing, and static pricing models, which result in slow response times, privacy vulnerabilities, billing errors, and queue bottlenecks.This project proposes **APMS v3.0**, an Automated Parking Management System that uses:

- FASTag-based RFID for seamless vehicle identification
- Edge AI for real-time occupancy detection
- Federated Learning for privacy-preserving License Plate Recognition
- Multi-Agent Reinforcement Learning (MARL) for dynamic pricing
- V2X geofencing for predictive vehicle arrival and slot allocation

Assumptions and Constraints

- Vehicles are equipped with FASTag for RFID-based identification.
- CCTV cameras and RFID readers are available at gate nodes.
- Campus-level infrastructure supports edge node deployment.
- Simulated campus environment reflects realistic parking congestion scenarios.

Stakeholders

- Vehicle Owners / Drivers
- University Administration
- Parking Management Staff
- Municipal Smart City Authorities
- SRM Institute of Science and Technology
- Project Guide – Dr. Harriet Linda C
- Department of Computing Technologies

Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Akshat Gupta

- Designed overall APMS architecture and system workflow
- Implemented FASTag RFID-based identification module
- Developed Flask backend and SQLite database setup
- Built baseline parking allocation and billing logic

Krish Nakul Gohel

- Designed Computer Vision pipeline for occupancy detection
- Implemented OpenCV ROI extraction and MOG2 background subtraction
- Developed slot availability monitoring system
- Created basic frontend dashboard UI

Week 1

Requirement analysis, parking workflow design, architecture planning

Week 2

Backend development, database schema creation, RFID flow design

Week 3

CV occupancy detection implementation and baseline testing

Week 4

Dashboard integration and baseline validation

Sprint Outcome

Initial APMS prototype completed with RFID identification, occupancy detection, slot monitoring, and billing workflow.

Signature of the Supervisor

Division of work and contributors of SPRINT 2 [Include Daily Scrum of SPRINT 2]

Sprint 2: Edge AI Optimization, Federated Learning and Dynamic Pricing

Duration: February 2026

Division of Work

Akshat Gupta

- Implemented Federated Learning framework for LPR privacy
- Designed simulated TensorRT/OpenVINO optimization layer
- Developed latency reduction pipeline

Krish Nakul Gohel

- Implemented MARL dynamic pricing engine
- Designed reward function and congestion threshold model
- Performed performance benchmarking and optimization

Daily Scrum Summary

Week 1

FL architecture design and local OCR training setup

Week 2

TensorRT optimization and latency testing

Week 3

MARL pricing engine implementation and Q-learning logic

Week 4

Performance evaluation and benchmark comparison

Sprint Outcome

Inference latency reduced from 45 ms to 12 ms with 3.75x speedup and predictive pricing stabilized occupancy during peak hours.

Signature of the Supervisor

Division of work and contributors of SPRINT 3 [Include Daily Scrum of Sprint 3]

Sprint 3: Blockchain Integration, V2X Geofencing and Final Deployment

Duration: March – April 2026

Division of Work

Akshat Gupta

- Implemented private blockchain ledger and smart contract billing
- Developed SHA-256 transaction hashing system
- Integrated V2X geofencing and 500m arrival prediction module

Krish Nakul Gohel

- Built mobile slot navigation using geotagged pins
- Developed audit dashboard and reporting module
- Conducted final testing, validation, and documentation

Daily Scrum Summary

Weeks 1–2

Blockchain integration and billing security validation

Weeks 3–4

V2X geofence implementation and slot navigation system

Weeks 5–6

Final UI polish, testing, result validation, and report writing

Sprint Outcome

Complete APMS v3.0 deployed with secure billing, predictive vehicle arrival, dynamic pricing, and full system evaluation.

Signature of the Supervisor

Worksheet / Data collection / Observation etc

1. Dataset

Simulated parking management dataset including:

- Vehicle entry and exit records
- Occupancy patterns
- FASTag vehicle identification
- Dynamic pricing decisions
- Arrival prediction events
- Parking duration and billing logs

2. Experimental Metrics

Measured:

- LPR inference latency
- Queue transaction time
- Vehicle throughput
- Blockchain overhead
- Geospatial slot assignment accuracy

3. Key Observations

- Standard OpenCV latency was 45 ms
- Optimized Edge AI latency reduced to 12 ms
- Queue time reduced from 5.8s to 3.5s
- Total throughput improved by 65%
- Blockchain overhead remained negligible (+45 ms)

- Predictive pricing prevented occupancy from reaching critical congestion thresholds

4. Tools and Technologies Used

- Python 3.x
- Flask
- SQLite (WAL-enabled)
- OpenCV
- TensorRT / OpenVINO simulation
- Federated Learning
- Blockchain (Private Ledger)
- SHA-256 hashing
- HTML, CSS, JavaScript
- Leaflet.js
- Git / GitHub

Research Article with Journal Publication Details / Patent disclosure form with patent status

(include certificates and proofs)

Research Article Details

Paper Title: APMS v3.0: A Decentralized Smart Parking Framework with Edge AI and Federated Learning

Authors: Akshat Gupta, Krish Nakul Gohel

Guide: Dr. Harriet Linda C
Assistant Professor, Department of Computing Technologies, SRM IST

Key Contributions

- Federated Learning for privacy-preserving License Plate Recognition
- Blockchain-enabled smart contract billing
- MARL dynamic pricing for congestion control
- Edge AI latency optimization
- V2X geofencing for predictive parking allocation

Performance Results

- 73.3% latency improvement
- 65% increase in throughput
- 100% slot assignment accuracy
- Secure immutable billing architecture

Publication Target

28/04/2026, 10:36 Conference Management Toolkit - Submission Summary

Submission Summary

Conference Name
Artificial Intelligence for Sustainable Energy Systems: Algorithms, Applications, and Impacts

Paper ID
24

Paper Title
APMS v3.0: A Decentralized Smart Parking Framework with Edge AI and Federated Learning

Abstract
This research proposes a comprehensive, multi-tiered intelligent framework for an Automated Parking Management System (APMS v3.0) designed to fundamentally address the escalating crisis of urban parking congestion, which currently accounts for an estimated 1.5 billion lost vehicle-hours annually in Indian metropolitan areas. While traditional parking solutions rely heavily on manual ticketing, reactive occupancy triggers, or highly vulnerable centralized cloud-based architectures, this study introduces a decentralized "Intelligent Edge" approach that leverages the existing National Electronic Toll Collection (NETC) FASTag infrastructure for seamless, contactless vehicle identification. The primary contribution of this work is the robust integration of five sophisticated computational modules into a unified Internet of Things (IoT) stack. To ensure absolute user privacy, a Federated Learning (FL) architecture is utilized for License Plate Recognition (LPR), allowing local gate nodes to train on sensitive vehicle data and share only aggregated model weighting parameters. To guarantee real-time responsiveness, a simulated TensorRT/OpenVINO optimization layer reduces computer-vision inference latency on low-power campus hardware. Financial transparency is secured through a private blockchain ledger and smart-contract-based billing engine that establishes an immutable audit trail for municipal revenue.

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28/04/2026, 10:36 Conference Management Toolkit - Submission Summary

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Primary Subject Area
Chapter 01 - AI in the Global Energy Transition: Opportunities and Challenges

Submission Files
920,021 Research_Paper_Final.pdf (2.3 Mb, 2026-04-28 10:34:02 GMT+5:30)

https://cmt3.research.microsoft.com/AISES2027/Submission/Summary/24 2/2

Current Status

Research paper completed and ready for conference submission.